# RM0016: printed off-ramp queue consistency check

Review note, 14 September 2026

**Scope.** Zhang et al., DOI 10.1109/MITS.2023.3323029, supplied final PDF p. 7 (printed p. 46), Eqs. (7)–(8). The equations were checked against a rendered page, not only extracted text. This note assesses the printed recurrence, not an unavailable implementation.

**Notation.** Write $Q_t = N_{j,t}^{\mathrm{OFF}}$ for off-ramp queue mass, $S = S_j^{\mathrm{OFF}}$ for storage, $d_{t+1} = q_{n-1,j,t+1}$ for arriving demand rate, $c = e_j^{\max}$ for downstream capacity rate, and $T > 0$ for step duration. The printed equations become

$$e_{t+1} = \min\left\{d_{t+1}, \frac{S - Q_t}{T}\right\}, \tag{1}$$

$$Q_{t+1} = \max\{Q_t + T(e_{t+1} - c), 0\}. \tag{2}$$

Assume fixed $S \geq 0$, $c \geq 0$, nonnegative demand and $0 \leq Q_t \leq S$. These are the ordinary feasible-state conditions, not an additional control constraint.

**One-step bound.** The first equation implies

$$Te_{t+1} \leq S - Q_t.$$

Consequently,

$$Q_{t+1} \leq \max\{S - Tc, 0\} \leq S.$$

Nonnegativity follows from the second equation. By induction, a feasible initial queue remains at or below storage. In particular, a trigger defined by $Q_t > S$ cannot activate through this recurrence from a feasible initial state.

**Why clarification is needed.** The article describes a lane-drop trigger when the off-ramp queue exceeds storage, and reports relative queues above one (e.g., Fig. 9). Those claims cannot be generated by the literal recurrence under the conditions above. Soft overflow penalties do not change this bound. A different plant update, different queue definition, changing storage, initially overflowing state, or printing error would need to be specified explicitly.

**Review consequence.** Do not silently replace the equations, infer misconduct, or use this discrepancy as a proof that ramp metering is infeasible. It blocks unqualified literal use of this source. Its conceptual role and the wider B1 feasibility question must be judged separately.